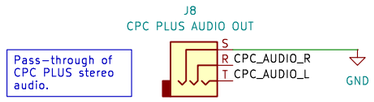
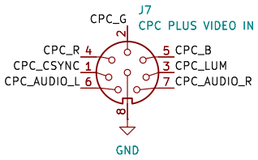
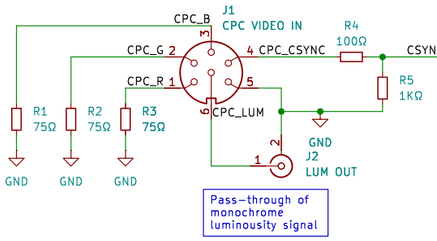


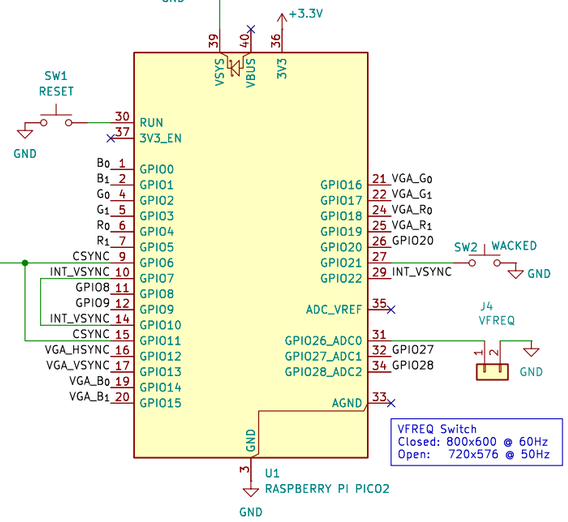
Pass-through of CPC PLUS stereo audio.



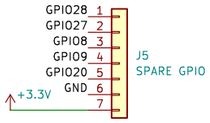


Pass-through of monochrome luminousity signal

Use 2.1mm splitter to power both CPC & Pico



VFREQ Switch
Closed: 800x600 @ 60Hz
Open: 720x576 @ 50Hz

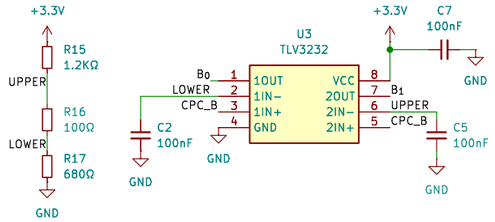


From CPC Video Connector to Comparators:
CPC_R
CPC_G
CPC_B

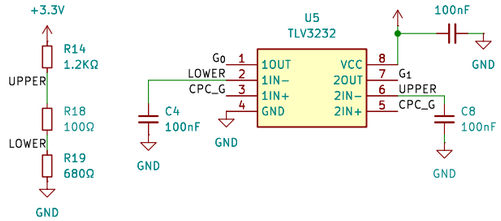
CPC_CSYNC converted to 3.3V CSYNC

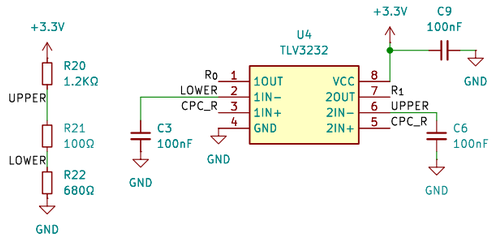
From comparators to Pico (2-bits per colour):
R0 & R1
G0 & G1
B0 & B1

Pass-through:
LUM (B&W video)
CPC PLUS audio



Duplicated resistor ladders to simplify PCB routing

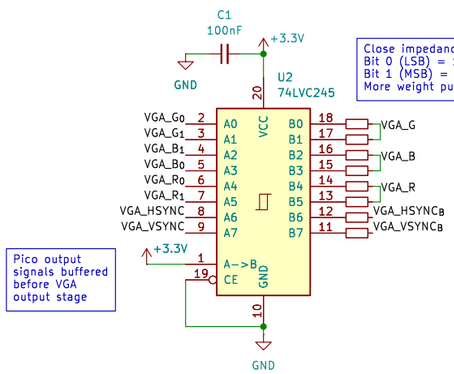




J3	KOBICONN 163-179PH-FX 2.1MM DC POWER JACK OR SIMILAR
J8	SAME SKY SJ3-35083A-TR 3.5MM STEREO JACK FOR CPC PLUS AUDIO
J1	SAME SKY SDJ-60J 6-PIN DIN FOR CPC VIDEO
J7	SAME SKY SDJ-80J 8-PIN DIN FOR CPC PLUS VIDEO
J4	2-PIN 2.54MM HEADER + STRAP
J6	AMPHENOL L77HDE15SD1CH4RHNVGA DE15 VGA OR SIMILAR
J2	KOBICONN 156-0099-E RCA PHONO JACK OR SIMILAR FOR LUM
J5	7-PIN 2.54MM HEADER (SPARE GPIO PORTS)
C	CERAMIC CAPACITOR P2.5MM
R	RESISTOR D1.6MM
SW1	TACTILE SWITCH 6MM
SW2	TACTILE SWITCH 6MM (SPARE SWITCH)
U1	RASPBERRY PI PICO2
U2	NEXPERIA 74LVC125AD-Q100J OR SIMILAR (SOIC-20)
U3	TEXAS INSTRUMENTS TLV3232QDGKRQ1 (SSOP-8)
U4	TEXAS INSTRUMENTS TLV3232QDGKRQ1 (SSOP-8)
U5	TEXAS INSTRUMENTS TLV3232QDGKRQ1 (SSOP-8)

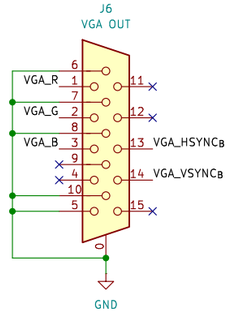
NOT YET TESTED

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole



Close impedance match to VGA:
Bit 0 (LSB) = 150Ω
Bit 1 (MSB) = 75Ω
More weight put on MSB.

Pico output signals buffered before VGA output stage



Based on original design by grzegorz-gr:
https://github.com/grzegorz-gr/vga4cpc
Enhanced firmware by WacKEDmaN:
https://github.com/WacKEDmaN/VGA4CPC-Enhanced
Brett Hallen

Sheet: /
File: VGA4CPC_Brett_RevE.kicad_sch

Title: CPC to VGA Converter

Size: A3	Date: 30/MAY/2026	**Rev: E**
KiCad E.D.A. 9.0.9		Id: 1/1